

MARITIME AI FRAMEWORK (MAF)

Comprehensive Technical Implementation Report

STATUS: Operational Integration Complete

DATABASE CONTEXT: Cassandra & Neo4j

KAFKA EVENT LOG: Real-time validated & anomalies

DATE: June 2026

Maritime AI Framework (MAF) - Comprehensive Technical Report

This report provides a detailed overview of the system architecture, data ingestion pipelines, database schemas, and analytical detection rules implemented in the Maritime AI Framework (MAF).

1. System Architecture & Data Flow

The Maritime AI Framework utilizes a real-time event-driven architecture coupled with a polyglot storage approach (Cassandra for high-throughput time-series telemetry + Neo4j for relational knowledge graphs).

```
graph TD
%% Ingestion Sources
A[AISStream.io WebSocket / Replay File] -->|Raw JSON| B(AIS Ingestor)
C[OFAC, UN, EU, OpenSanctions Feeds] -->|XML / JSON| D(Sanctions Ingestor)

%% Messaging
B -->|ais.validated| K_VAL[Kafka: ais.validated]

%% Analytics & Analysis
K_VAL -->|Consume| E(Signal Analyser)
E -->|Publish Anomalies| K_ANOM[Kafka: ais.anomalies]

%% Cassandra Storage Layer
K_VAL -->|Consume| F(AIS History Writer)
F -->|Write Position / 90d TTL| C_POS[(Cassandra: ais_positions)]
F -->|Upsert Last Coordinates| C_SUM[(Cassandra: vessel_track_summary)]
F -->|Log Silence Gaps| C_DARK[(Cassandra: dark_event_candidates)]
F -->|Increment Counters| C_STATS[(Cassandra: dashboard_stats)]

%% Graph Storage Layer
K_VAL -->|Consume| G(Neo4j ETL)
K_ANOM -->|Consume| G
D -->|Write Sanctions & Alerts| N_DB[(Neo4j: Knowledge Graph)]
G -->|Upsert Vessels / Flags / Events| N_DB

%% Presentation Layer
N_DB -->|Cypher Queries| H(FastAPI / Flask Server)
C_POS -->|Track History Queries| H
C_STATS -->|Counter Point-Lookups| H
H -->|Serve Dashboard API| I[Leaflet.js UI Dashboard]
```

2. Data Ingestion Pipelines & Sources

The platform processes five distinct categories of data to provide a comprehensive, multi-dimensional security and tracking profile:

2.1 Live AIS Telemetry

- Source: WebSockets from AISStream.io providing global live AIS broadcasts.
- Ingestion Handler (ingestor.py):
- Connects to the streaming WebSocket using API credentials.
- Replay Mode: Falls back to parsing a local 45,000-record dataset (sample_ais.ndjson) to simulate live streaming during testing.
- Supported Messages: PositionReport, StandardClassBPositionReport, and ShipStaticData.
- Validation Engine: Filters out invalid coordinates (lat outside +/-90, lon outside +/-180) and physically impossible speeds (>102.3 kts).
- Output: Validated reports are formatted as uniform JSON envelopes and published to the ais.validated Kafka topic.

2.2 Sanctions Ingestion & Entity Resolution

- Source: Four international regulatory feeds updated every 15 minutes by the sanctions_ingestor.py cron service:
 1. OFAC SDN List (US Department of the Treasury) - XML format.
 2. UN Consolidated List (UN Security Council) - XML format.
 3. EU Financial Sanctions File (FSF) (European Commission) - XML format.
 4. OpenSanctions Vessels Export - Line-delimited JSON format.
- Fault-Tolerance: Automatically falls back to locally cached XML files if a network timeout occurs. If any feed fails 3 consecutive times, it merges a active SanctionsFeedAlert node into Neo4j to trigger an operator alert.
- Entity Resolution & Deduplication:
- Elements like MMSI, IMO number, call signs, and flag states are extracted.
- Entities are merged and deduplicated by matching IMO Numbers.
- Deduplicated profiles merge their sanction sources and program designations (e.g. merging OFAC-IRAN-01 and EU-FSF profiles).
- Merged entries are written into the Neo4j graph database using an optimized UNWIND Cypher batch query.

2.3 Maritime Registry Web Scraping & Cross-Referencing

- Sources: Direct scraping selectors for international maritime registries:
 1. MarineTraffic AIS Vessel Details - Live HTML scraping using BeautifulSoup parsers.
 2. Equasis Restricted Ship List - Secure authenticated sessions querying restricted registry records.
- Ingestion Handler (scrapers.py):
- Concurrent Scraping: Orchestrates parallel requests using a RegistryCrossReferencer built on ThreadPoolExecutor to minimize request latencies.
- Deduplicated Fields: Extracts ship owner names, operating company IMOs, vessel name history, and Port State Control (PSC) inspection/detention statuses.
- Resiliency & Fallbacks: Features a mock database fallback mapping (e.g., resolving IMO 9988776 to National Iranian Tanker Co and SHADOW FLEET CORP) if Cloudflare bot protections or login Captchas block live session

requests.

2.4 Geospatial EEZ Shapefiles

- Source: Bounding polygon coordinates of maritime zones from the Marine Regions World EEZ v12 Shapefile (World EEZ database).
- Ingestion Handler (eez_loader.py):
- Iterates and parses .shp and .shx records using PyShp and Shapely.
- Extracts the Marine Regions ID (MRGID), sovereign territory name, and country ISO codes.
- WKT Polygon Compression: Translates geometries to WKT (Well-Known Text). Polygons exceeding 5,000 characters are simplified using Shapely's coordinate reduction algorithms (simplify(0.05)) or capped using their bounding boxes (POLYGON) to maintain high-throughput indexing in Neo4j.
- Writes EEZ boundaries once at system startup to populate spatial check geometry nodes.

2.5 Authoritative Bathymetric Ocean Depth Grids

- Source: Authorized ocean floor coordinate matrices from the GEBCO Bathymetric Grids (General Bathymetric Chart of the Oceans).
- Ingestion Handler (bathymetry.py):
- Accesses authoritative gridded elevation datasets in NetCDF format (GEBCO_2026.nc) via Scientific Python libraries (Scipy and netCDF4).
- Bilinear Keel Clearance Lookup: Resolves geographic coordinates directly to grid indices and extracts seabed altitude (depth expressed as negative values).
- Trade-lane Spatial Model Fallback: If NetCDF datasets are missing, it resolves depths against coordinate bounds of major shallow shipping lanes (Strait of Hormuz, Strait of Malacca, Suez Canal, English Channel) with realistic geographic mathematical models.

3. Database Architecture (Polyglot Storage)

3.1 Cassandra: High-Volume Telemetry & Auditing

Designed to handle millions of incoming position reports without read degradation or unbounded disk usage.

- ais_positions Table:
- Primary Key: (mmsi, date_bucket, timestamp)
- Partitioning by MMSI and a daily date bucket keeps individual partition sizes small (approx. 144 rows for a standard vessel pinging 6 times per hour).
- Clustering by timestamp DESC guarantees that track histories and latest pings are retrieved without an expensive sort phase.
- Automatic Expiry (TTL): Written with a 90-day TTL (7,776,000 seconds). Compaction uses the TimeWindowCompactionStrategy (TWCS, window = 1 day), enabling Cassandra to drop entire expired day-files at

once rather than scanning individual rows.

- vessel_track_summary Table:
- Keys on mmsi (Primary Key). Holds only the persistent, latest-known state (position, heading, SOG, identity) of each vessel. No TTL is applied.
- dashboard_stats Counter Table:
- Designed as a dedicated counter table (metric text PRIMARY KEY, value counter) to track total_positions and tracked_vessels.
- The history writer increments these counters asynchronously on every write.
- Bypasses expensive SELECT COUNT(*) queries on the telemetry tables, which hit Cassandra read timeouts due to scanning deleted/expired tombstones.
- Asynchronous Telemetry Ingestion (AsyncWriter):
- Utilizes Cassandra's asynchronous execute interface (execute_async) to queue concurrent writes, avoiding write blockages on the main ingestion thread.
- Regulates the pipeline using a configurable concurrency boundary (MAX_IN_FLIGHT = 500) and drains completed query futures on separate worker threads. Flushes pending queries on idle and commits Kafka offsets periodically.
- Batch Capacity: The Docker Compose service configuration tunes the batch size limit from 20 to 200 telemetry records to maximize parallel writes under this async framework.

3.2 Neo4j: Relation & Context Knowledge Graph

Maintains high-value contextual relationships to enable semantic querying and spatial pattern analysis.

- Schema Nodes:
- (v:Vessel): Holds MMSI, IMO, vessel name, and running risk scores.
- (f:Flag): Represents flag states (linked via [:FLAGGED_UNDER] and [:REGISTERED_UNDER]).
- (s:Sanction): Represents regulatory programs.
- (e:Event): Captures anomalies (e.g., loitering, AIS gaps, beacon cadences).
- Indexing:
- Unique constraints enforce ID uniqueness for fast lookups.
- Neo4j Point Indexes are defined on v.location and e.location to accelerate geospatial proximity queries (e.g. proximity to ports or EEZs).
- Sanctions Match Synchronization: A periodic query matches (v:Vessel) and (s:Sanction) by IMO number, creating [:SANCTIONED_BY] relationships and flagging the vessel as sanctioned = true.
- Container Race Prevention:
- Implements connection-retry verification blocks in Neo4j-dependent ingestion elements (eez_loader.py and signal_analyser.py).
- Services attempt graph connections up to 20 times (sleeping 10 seconds between attempts) to gracefully await database startup, schema generation, and index configuration.

4. Anomaly & Dark Vessel Detection Rules

The framework applies specific physical, statistical, and spatial rules to identify suspicious activity.

Code Name	Category	Trigger Condition
M-AIS-BEACON	Cadence Anomaly	Coefficient of Variation (CV) < 0.01
M-SPEED-ANOMALY	Velocity Violation	SOG > allotted max speed per vessel type (e.g. Tanker > 20 kts)
M-EEZ-VIOLATION	Spatial Boundary	Vessel Flag State != Sovereign EEZ Country
M-DARK-EVENT	Telemetry Outage	Silence duration >= 2.0 hours
M-TRACK-PLAUSIBILITY	Kinetic Plausibility	Draft > bathymetric depth OR sharp turn (>120 deg) at speed (>10 kts)
M-LOITERING	Behavior Anomaly	Speed = 12.0 hours in open water
M-MMSI-SPOOF	Identity Spoofing	>2 distinct vessel names broadcasting the same MMSI
M-FLAG-HOP	Identity Evasion	>2 flag state changes within a short registration period
M-IDENTITY-LAUUNDER	Identity Obfuscation	IMO persists while MMSI, name, and flag change simultaneously
M-IMO-CLONE	Identity Cloning	Multiple distinct MMSIs claiming the identical IMO number
M-PORT-CALL-ANOMALY	Voyage Pattern	Irregular/unexpected port visits in trade sequences
M-SANCTIONED-ZONE	Spatial Regulatory	Entry into sanctioned port/waters or flag of sanctioned state
M-STS-TRANSFER	Rendezvous Risk	Co-loitering/loitering events close in space and time with another vessel
M-FLEET-BROKER	Corporate Link	Owned/managed by broker connected to >5 other vessels with sanctioned ties
M-SHELL-CHAIN	Corporate Opacity	Deeply layered corporate ownership chain (depth > 3 levels)
M-SISTER-RISK	Risk Propagation	Structurally identical sister ship under active sanctions

4.1 M-AIS-BEACON (Cadence Regularity)

- Logic: AIS transponders on moving ships transmit at irregular intervals based on speed and rate-of-turn. Unnaturally constant intervals suggest simulated, scripted mock telemetry.
- Math: The `signal_analyser.py` tracks a sliding window of the last 20 pings for each MMSI. It computes the Coefficient of Variation (CV) of the time difference between consecutive pings:

$$CV = \text{standard_deviation} / \text{mean} \quad (CV = \sigma / \mu)$$

where sigma is the standard deviation of intervals and mu is the mean interval.

- Threshold: If $CV < 0.01$, the variance in the cadence is extremely small ($<1\%$). It flags the Cadence as an automated beacon pattern.

4.2 M-SPEED-ANOMALY (Velocity Violation)

- Logic: Flags telemetry where a vessel's Speed Over Ground (SOG) exceeds logical physical limits allotted for its specific vessel type.
- Thresholds:
- Fishing / Tug / Towing / Dredging: 15.0 knots

- Tanker: \$20.0\text{ knots}\$
- Sailing: \$20.0\text{ knots}\$
- Pleasure Craft / Cargo: \$25.0\text{ knots}\$
- Passenger: \$30.0\text{ knots}\$
- High-Speed Craft (HSC): \$45.0\text{ knots}\$
- Default / Other: \$35.0\text{ knots}\$
- Confidence Formula:

Confidence = $\min(1.0, (\text{SOG} - \text{Threshold}) / \text{Threshold})$

4.3 M-EEZ-VIOLATION (Geospatial Entry Check)

- Logic: Detects foreign-flagged commercial vessels entering sovereign Exclusive Economic Zones (EEZ) without authorization.
- Execution:
- Imports sovereign EEZ boundaries as GeoJSON/WKT polygons from Neo4j once at startup.
- Conducts a real-time shapely point-in-polygon check for each incoming coordinate.
- If a vessel enters an EEZ and its registered flag (ISO Alpha-2 code) does not match the sovereign country ISO code of the zone, the framework flags an unauthorized entry.
- State is tracked in-memory to limit the alert to a single trigger upon crossing the boundary.

4.4 M-DARK-EVENT (AIS Outage / Silence)

- Logic: Flags a vessel when it disables its transponder to mask its movements (e.g. prior to oil smuggling, ship-to-ship rendezvous, or entry into conflict zones).
- Execution:
- The DarkEventTracker runs in-memory within the history writer, storing last_seen timestamps.
- Every 60 seconds, it evaluates all vessels. If the difference between the current time and a vessel's last received transmission is ≥ 2.0 hours, it logs a candidate dark event.
- A record is written to the dark_event_candidates table in Cassandra including the last known coordinates, the time the gap began, and the silence duration.
- Reappearance: When the vessel reappears and sends a new telemetry message, the tracker removes it from the silent list, allowing it to be evaluated again for future outages.

4.5 M-TRACK-PLAUSIBILITY (Kinetic Plausibility & Depth Verification)

- Logic: Detects physical impossibility in vessel movements or draft requirements.
- Verification:
- Geospatial Depth: Checks the vessel's current draft against the bathymetric depth at its current lat/lon (using the BathymetryEngine). If the draft is greater than or equal to the water depth, it indicates a depth violation or spoofed position telemetry.
- Kinetic Plausibility: Evaluates the heading difference between consecutive points. If the vessel undergoes a sharp turn ($>120^\circ$) while maintaining a speed $>10\text{ knots}$, it is flagged.

4.6 M-LOITERING (Suspicious Loitering)

- Logic: Identifies vessels hovering or drifting in open waters for extended periods outside designated anchorages or fishing grounds, typical of Ship-to-Ship (STS) transfer prep or smuggling.
- Threshold: Speed stays below 2.0 knots for a duration of at least 12.0 hours .

4.7 M-MMSI-SPOOF (MMSI Identity Spoofing)

- Logic: Identifies cases where multiple physically distinct vessel names broadcast using the same MMSI, or a single MMSI shows physically impossible simultaneous positions (telemetry collision).
- Trigger: More than 2 distinct vessel names claimed by the same MMSI.

4.8 M-FLAG-HOP (Flag Registration Evasion)

- Logic: Flags vessels that register under multiple different flag states in a short time frame. This is a common evasion technique to bypass flag-specific sanctions or inspections.
- Trigger: More than 2 flag changes within a year for a stable IMO identifier.

4.9 M-IDENTITY-LAUUNDER (Identity Laundering)

- Logic: Detects cases where the IMO number remains constant, but the MMSI, vessel name, and flag all change simultaneously to sever historical audit trails.
- Trigger: Simultaneous modification of MMSI, vessel name, and flag registry state.

4.10 M-IMO-CLONE (IMO Cloning)

- Logic: Identifies when two or more distinct vessels (using different MMSI identifiers) claim the same IMO number in overlapping time windows.
- Trigger: Multiple active MMSIs associated with a single IMO number in Neo4j.

4.11 M-PORT-CALL-ANOMALY (Port Call Sequence Violation)

- Logic: Flags statistically irregular sequences of port visits relative to the vessel's trade class or historical trading patterns.
- Trigger: Voyage routing deviations violating standard route probability graphs.

4.12 M-SANCTIONED-ZONE (Restricted Operations)

- Logic: Flags vessels entering restricted ports/waters or operating under the flag of sanctioned states.
- Trigger: Entering active EU/OFAC exclusion zones, or registered under high-risk flags (e.g. IR, KP, RU).

4.13 M-STTS-TRANSFER (Ship-to-Ship Transfer Risk)

- Logic: Detects two vessels co-loitering in close proximity in open seas to transfer cargo without customs

inspections.

- Trigger: Temporal and spatial overlap of loitering or dark events between two vessels.

4.14 M-FLEET-BROKER (Fleet Broker Connection)

- Logic: Identifies vessels managed by a central corporate entity showing high betweenness centrality and managing multiple blacklisted or sanctioned vessels.
- Trigger: Connection to a manager representing more than 5 vessels with sanction records.

4.15 M-SHELL-CHAIN (Obfuscated Ownership)

- Logic: Identifies deeply layered corporate shell ownership structures designed to hide the ultimate beneficial owner (UBO).
- Trigger: Ownership structure exceeding 3 levels of corporate subsidiaries removed from the operating entity.

4.16 M-SISTER-RISK (Sister Ship Risk Propagation)

- Logic: Propagates risk scores to structurally identical sister ships (sharing build specs, shipyards, or parent company) of blacklisted vessels.
- Trigger: An active sister vessel inherits risk propagation due to direct association with a sanctioned vessel profile.

5. Presentation Layer & Dashboard

- Flask Dashboard Server (dashboard_server.py):
- Integrates endpoints to feed the Leaflet.js map and control widgets.
- /api/cassandra: Pulls metrics from the dashboard_stats counter table to display the total messages and tracked vessels on the UI in less than 50 milliseconds.
- /api/vessel-tracks/: Queries Cassandra across the daily buckets of the last 90 days (adjustable from 1 to 90 days via the frontend history slider) to plot the historical path of a selected vessel on the Leaflet map.
- /api/vessel-path/: Resolves complex relationships in Neo4j (FlagStates, corporate operators, sanction lists, and linked reports) to deliver structured JSON details on demand.
- Operations Geospatial Map UI (maritime_map_osint_dashboard.html):
- Interactive GIS Map: A dark-mode Leaflet map using CartoDB Dark Matter tile layer, centered on the high-risk Arabian Sea / Indian Ocean shipping corridor.
- Dynamic Marker Styling:
- Normal Vessels: Cyan circle markers with a glowing shadow.
- Anomalous Vessels: Red pulsing markers (pulse-red CSS animation) to flag immediate kinetic or logical warnings.
- Silent/Dark Vessels: Yellow pulsing markers (pulse-warn CSS animation) designating active AIS outages.
- Multi-Dimensional Filtration Sidebar:
- Real-time Search: Real-time filtering of markers by MMSI or Vessel Name.

- Anomaly Checkboxes: Toggles map visibility for Speed Anomalies, Dark Events/AIS Gaps, EEZ Violations, Sanctioned Vessels, and Beacon Patterns.
- Flag State Selector: Dropdown to filter markers by specific flag countries.
- Speed Boundaries: Dual range sliders to filter vessel speeds between 0 and 99+ knots.
- Show Anomalies Only Toggle: Hides all non-flagged commercial traffic.
- Historical Track Reconstruction (1-90 Days):
 - Adjustable tracking duration slider linked directly to Cassandra daily date buckets (from 1 to 90 days).
 - Plots historical paths as dashed cyan vector polylines, marks the oldest track start, and places waypoint dots every 20 positions to represent speed variations.
 - Fits the Leaflet viewport boundaries automatically to the bounds of the loaded track.
- Vessel Detail Slide Panel:
 - Slides up from the bottom of the viewport upon selecting a vessel.
 - Displays identity badges (flag and vessel type code labels), alert warning strips, and a statistics grid (Speed, Heading, Draft, Position, and Last Seen).
 - Features a click-to-copy utility on the MMSI identifier to copy coordinates to the clipboard.
- Neo4j Intelligence Panel:
 - Queries `/api/vessel-path/` on-demand to display detailed identity metadata, nested parent company owner structures (company IMO), active sanction authority programs, and LLM verdicts/confidence scores.
- Diagnostics & Recovery Tools:
 - Status bar indicating connection state (CONNECTING..., LIVE, MOCK DATA) and a real-time UTC clock.
 - Health monitoring grid checking the health of Kafka, Cassandra, Neo4j, ZooKeeper, and active ingestion scripts.
 - ETL offset recovery banner allowing operator-guided offset reset (`/api/etl_reset_anomalies`) to resolve message queues.
- Stand-alone Demo Mode:
 - Automatically injects 250 mock vessels and 5 synthetic alerts if the backend server is unreachable, maintaining full interactivity for demonstration purposes.